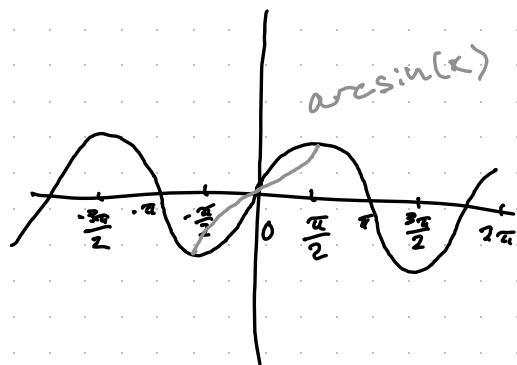


Inverse trig functions - 0.1.4



$\sin(x)$

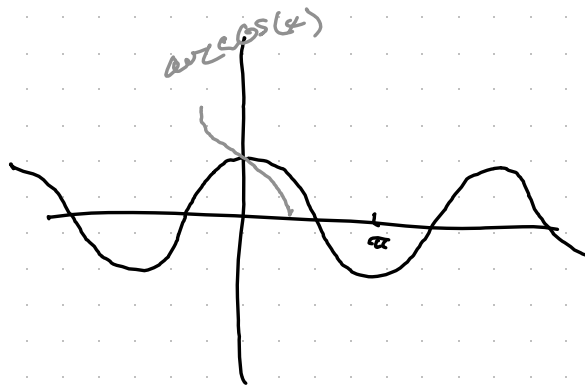
Work on $[-\frac{\pi}{2}, \frac{\pi}{2}]$

w/ range $[-1, 1]$

Inverse function has
domain $[-1, 1]$ and

range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

Call the inverse $\arcsin(x)$



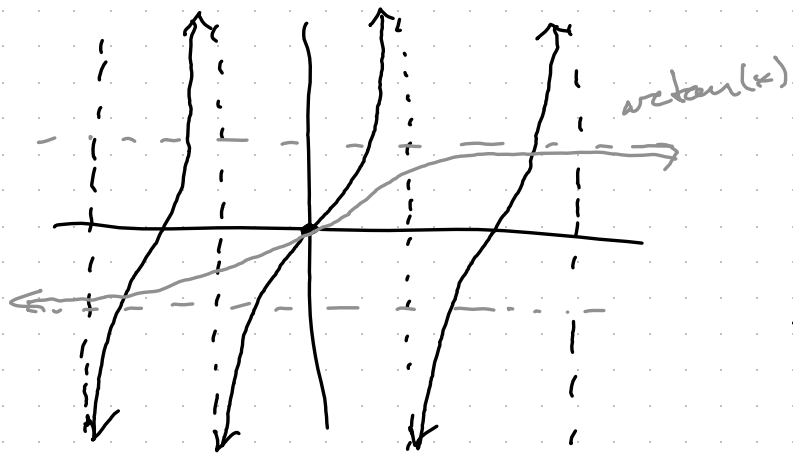
$\cos(x)$

Work on $[0, \pi]$ w/

range $[-1, 1]$

Inverse function has domain
 $[-1, 1]$ and range $[0, \pi]$

Call the inverse $\arccos(x)$



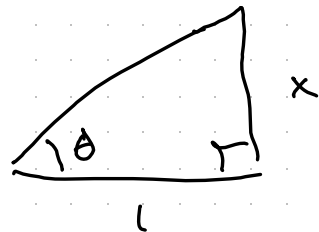
$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

Works on $(-\frac{\pi}{2}, \frac{\pi}{2})$
range is $(-\infty, \infty)$,

so $\arctan(x)$ has
domain $(-\infty, \infty)$ and
range $(-\frac{\pi}{2}, \frac{\pi}{2})$.

$\arctan(x)$ is the unique input to $\tan$
with output x .

Think of x as some ratio of side lengths



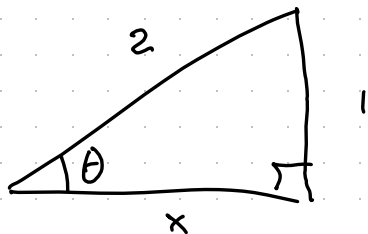
$$\arctan\left(\frac{x}{1}\right) = \theta$$

we get back the unique angle
between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ with tangent x .

Example: Compute $\tan(\arcsin(\frac{1}{2}))$.

$\underbrace{\hspace{1.5cm}}_{\theta}$

Input to $\arcsin$ is a ratio of side lengths



$$\tan \theta = \frac{\text{opp.}}{\text{adj.}}$$

Pythagorean theorem says $1^2 + x^2 = 2^2$

$$x^2 = 4 - 1 = 3 \Rightarrow x = \sqrt{3}$$

$$\tan(\arcsin(\frac{1}{2})) = \frac{1}{\sqrt{3}}$$

$$\rightarrow \tan(\arcsin(x)) = \frac{x}{\sqrt{1-x^2}}$$

